Partisan disparities in the use of science in policy

Documents from Congress and think tanks reflect differences in how science is cited

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[[TODO: presenter · event · date]]

Science informs policy on our hardest problems, even as politics polarizes

Climate change, public health crises, technological advance

The question has three parts: how much, which science, and why

1

How much?

Do the parties differ in how often they cite science?

2

Which science?

When they do cite it, is it the same science?

3

Why?

What might explain the differences?

We linked 641,894 policy documents to 122 million papers

OVERTON

641,894

US policy documents, 1995–2021

Federal government and think tanks

With the papers they cite

matched by DOI

DIMENSIONS

122

million scientific publications

All disciplines

Field, citation impact, preprint status

Congress and think tanks give two ways to read “party”

Congressional committees

Party is whichever party controls the committee.

Democratic-controlled

Republican-controlled

49,345 reports and hearings

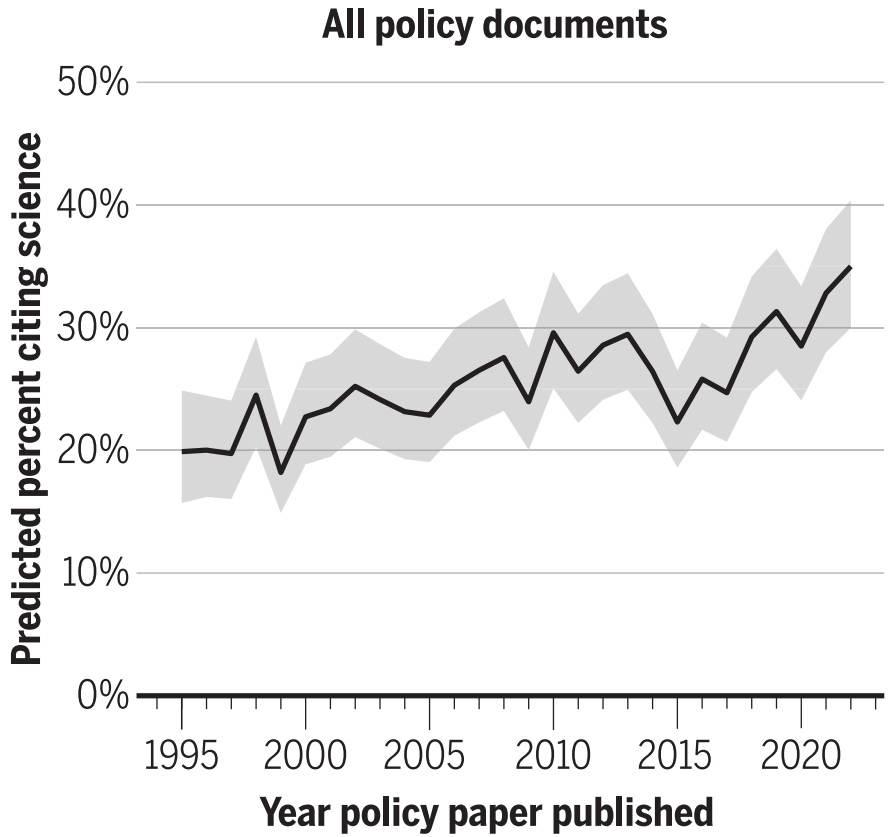
Ideological think tanks

Party is the think tank’s left-right position, coded from ideology scores and mission statements.

Left-leaning **Right-leaning**

121 think tanks · 191,118 documents

The share of policy documents citing science rose from under 20% to over 35%



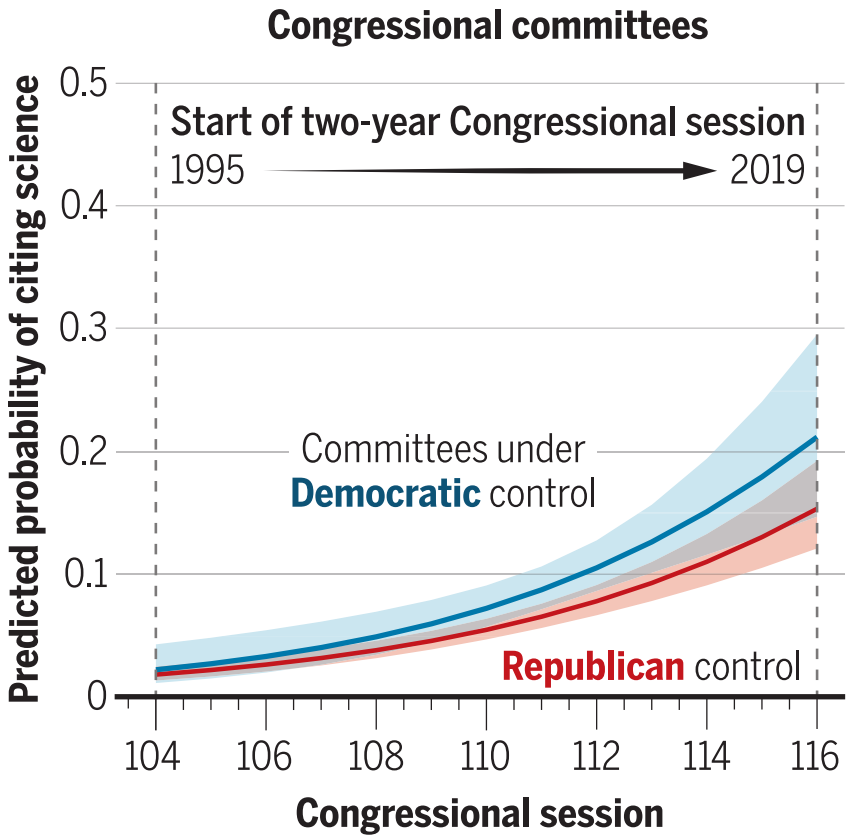
Band: 95% bootstrapped confidence interval

>35%

of policy documents cite science in 2020, up from under 20% in 1995

Both parties' documents are in this curve. Next: do they differ?

Democratic committees are 1.8 times more likely to cite science



Bands: 95% bootstrapped confidence intervals

1.8×

more likely to cite science when a committee is under Democratic control

Holds across chamber, document type, and whether the committee shares the President's party.

Switching a committee to Democratic control adds about 196 citations

+196

citations to science per committee in the Congress after control switched from Republican to Democratic

- Difference-in-differences on 80 committees across 13 Congresses
- An increase of 0.58 standard deviations
- No equivalent effect in placebo tests

Left-leaning think tanks are more than five times as likely to cite science



Bands: 95% bootstrapped confidence intervals

>5×

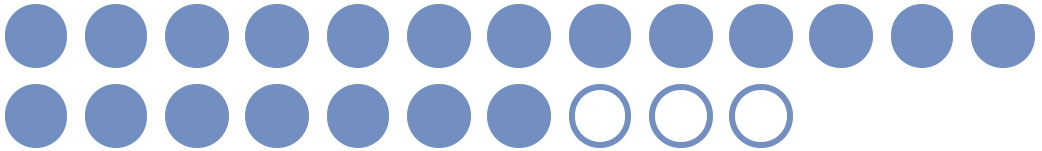
more likely to cite science than right-leaning think tanks

Think tanks' ideology is mostly static, so this is a comparison across organizations, not a switch.

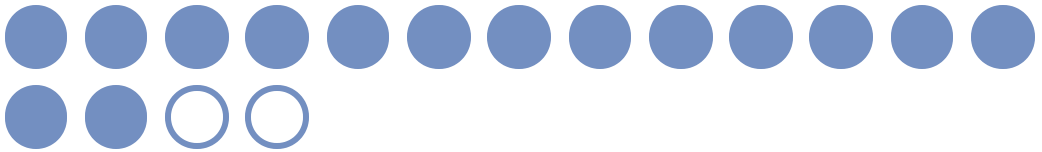
The gap appears in nearly every field and issue area

Democratic vs Republican committees

20 of 23 scientific fields

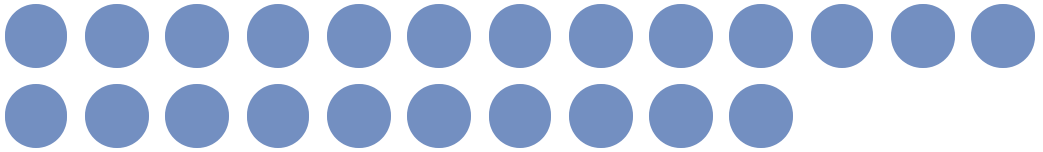


15 of 17 issue areas

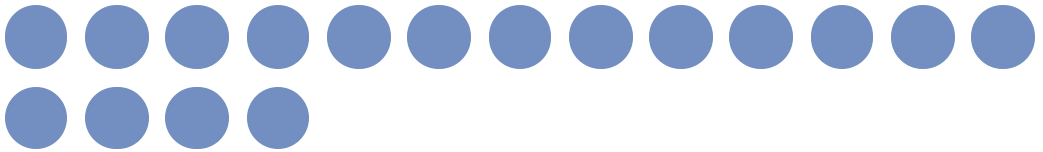


Left- vs right-leaning think tanks

23 of 23 scientific fields



17 of 17 issue areas



● Cites distinguishably more science ○ Not distinguishable

Only 5–6% of cited papers are cited by both parties

5–6%

of scientific citations in policy documents are shared by Republicans and Democrats

Partly because a paper cited only once cannot, by construction, be cited by both sides.

Chance alone would predict more than twice as much overlap

Observed

5–6%

of papers cited by both parties

vs

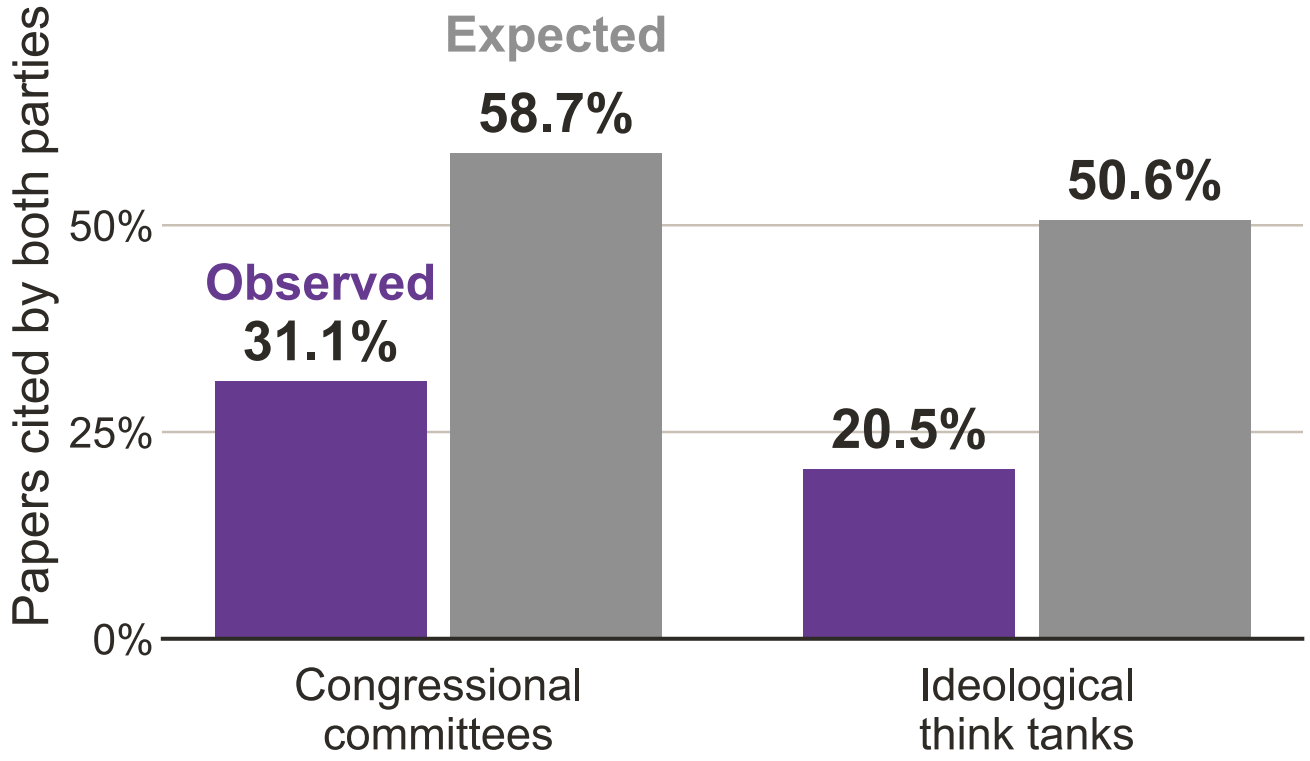
Expected

12–13%

if citing were uncorrelated with party

Expected value from a permutation test that keeps each paper's citations and each party's totals fixed.

Even papers that could be shared are shared far less than expected

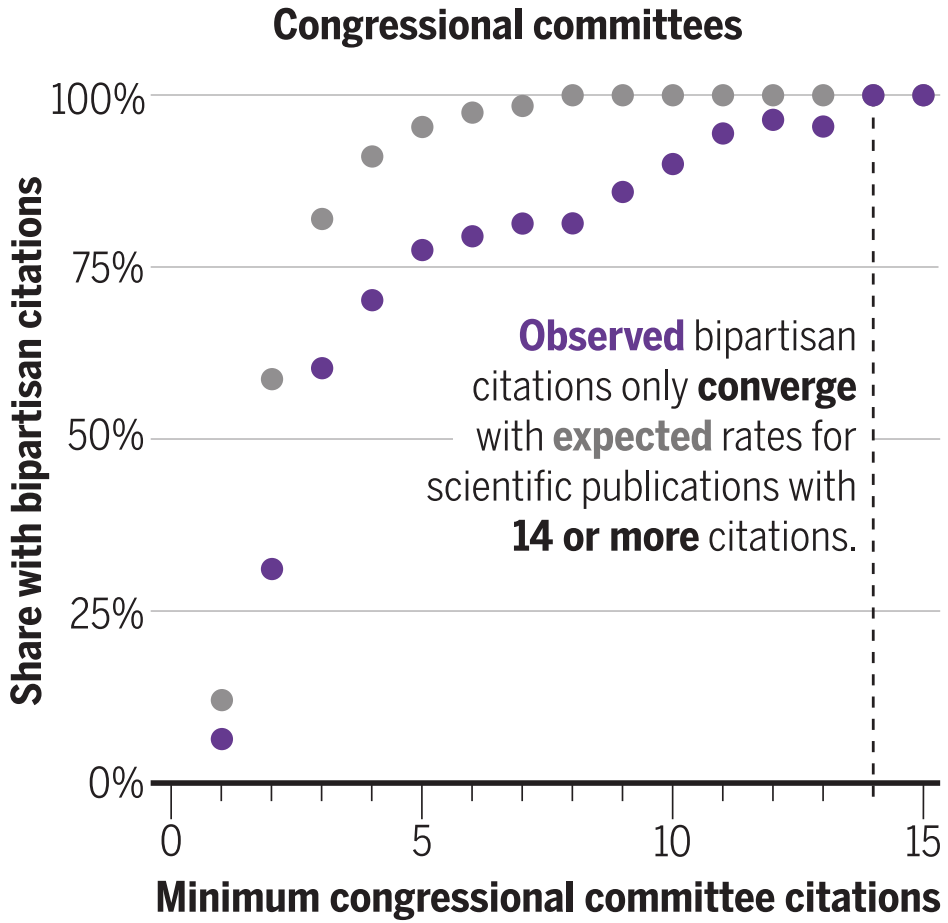


Conservative test

Only papers cited two or more times, the ones that could have been cited by both sides.

Permutation baseline preserves each paper's citation count and each party's totals

In Congress, overlap reaches expectation only for papers cited 14 or more times



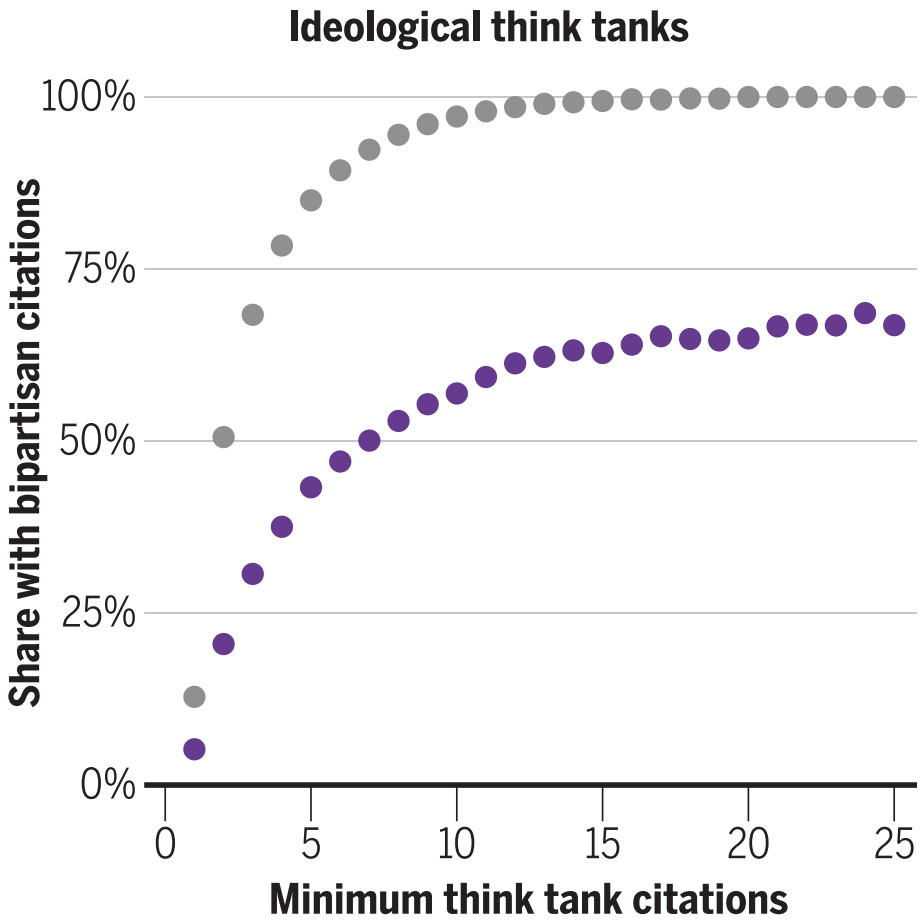
14+

citations before observed overlap matches the expected rate

14 citations is the top 99.9th percentile of policy citations.

● Observed ● Expected

Among think tanks, overlap never reaches expectation, at any threshold



<70%

of the expected rate: where observed overlap plateaus, above 20 policy citations

Even the most widely cited science is rarely cited by both sides.

Observed Expected

The low overlap is stable over time and shows up across fields and issues

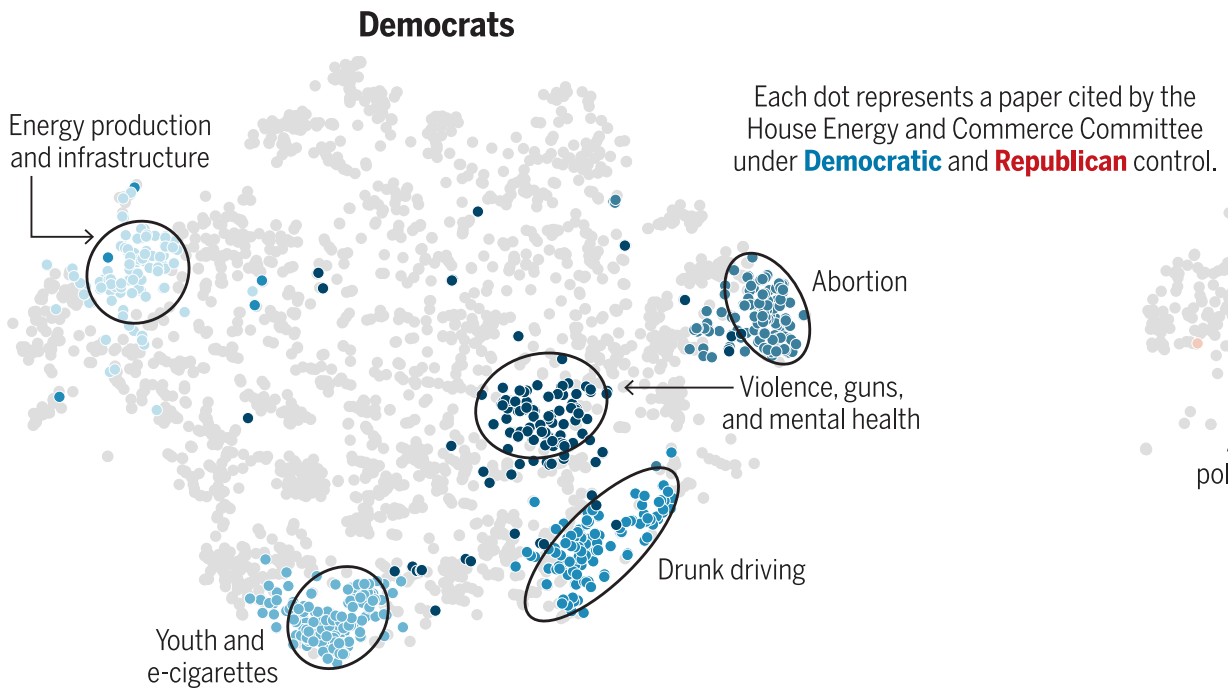
Over time

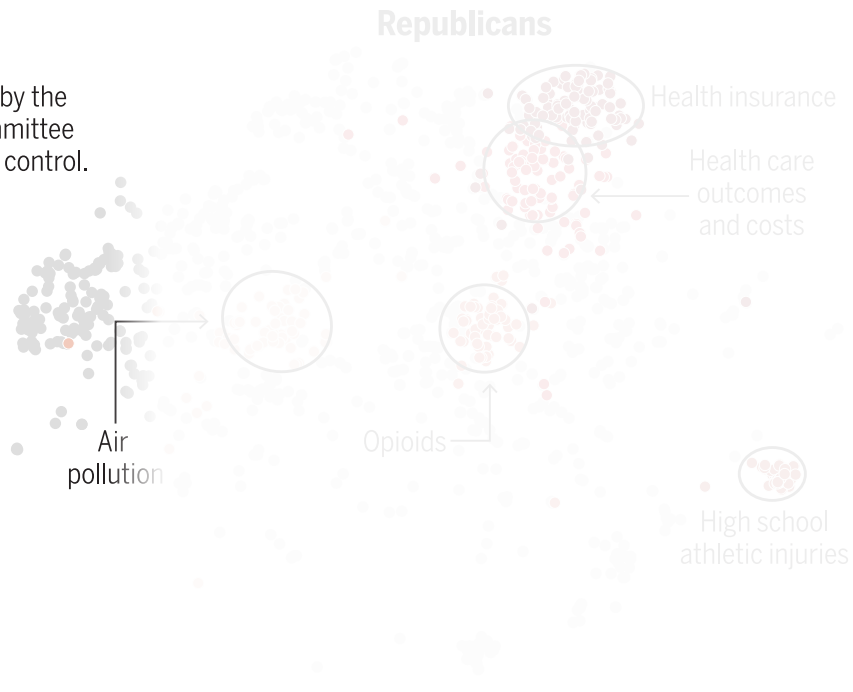
The degree of overlap has stayed rather stable, a long-standing feature of the US policy landscape.

Across the board

Low across scientific fields and policy issues, in Congress and in think tanks.

The same House committee cites different clusters of science under each party

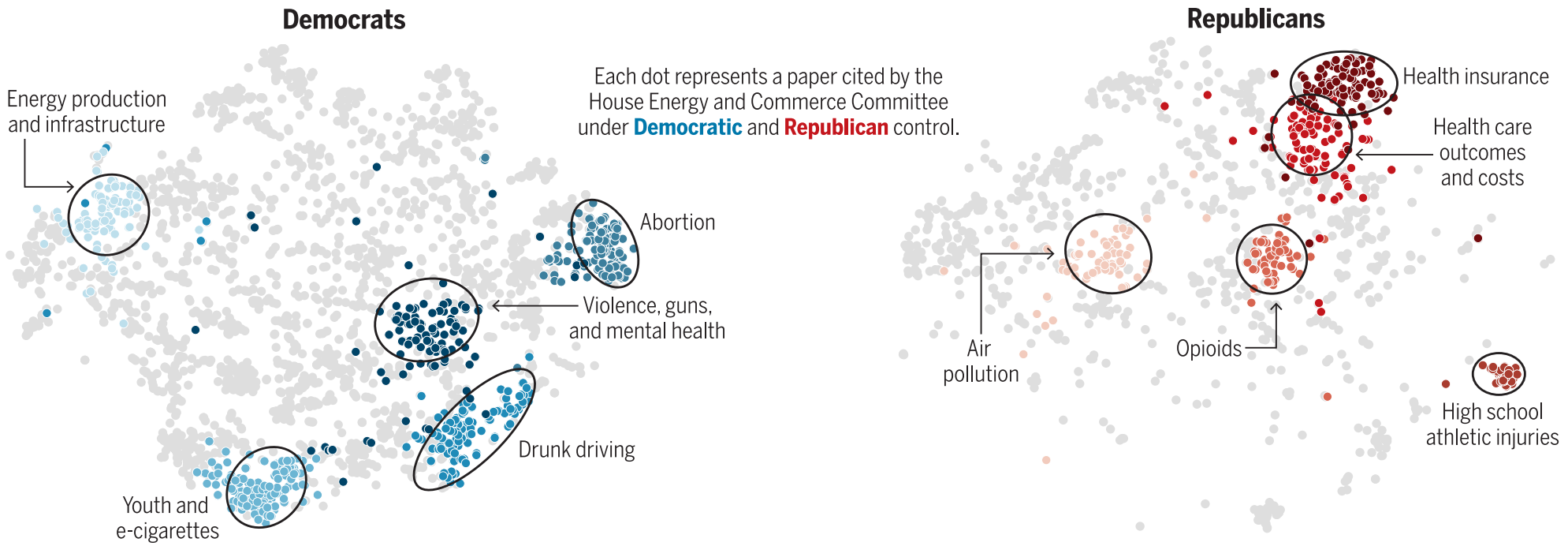




Democrats: abortion, drunk driving, youth and e-cigarettes

Republicans: health insurance, opioids, high school athletic injuries

The same House committee cites different clusters of science under each party



Democrats: abortion, drunk driving, youth and e-cigarettes

Republicans: health insurance, opioids, high school athletic injuries

Different agendas do not explain it: same-issue documents cite different papers

Urban Institute

left of center

Cites studies of distributional effects.

Employment Policies Institute

right of center

Cites studies of employment and labor-supply effects.

1 of 62 cited papers are shared by the two documents

Minimum-wage documents matched on near-identical titles

Same-party documents cite science about 0.1 closer than opposing-party documents

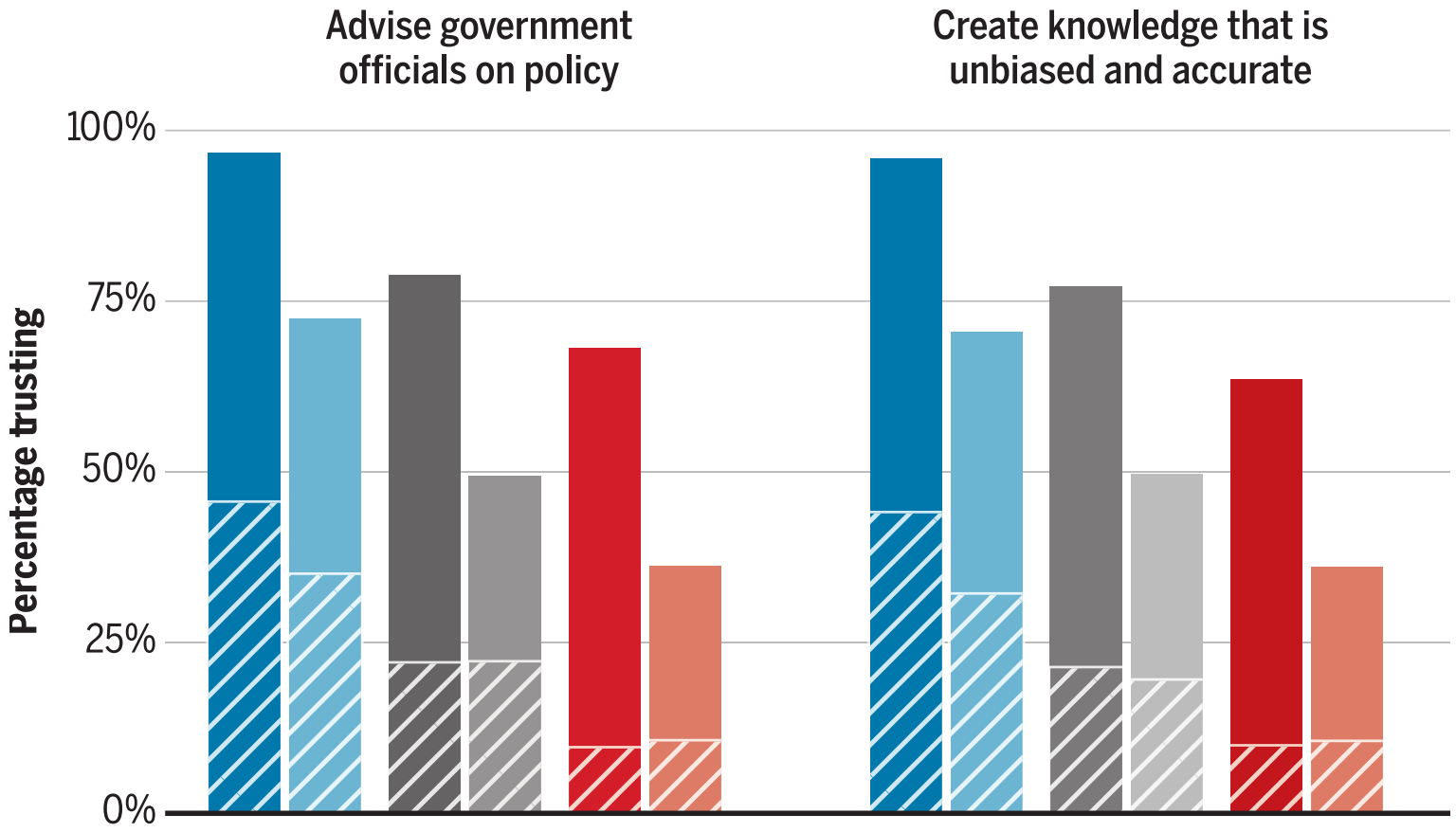
≈0.1

closer in cosine similarity between the papers two documents cite (0–1 scale)



Matched documents on the same issue, in both Congress and think tanks.

Political elites trust scientists more than likely voters do, in every party



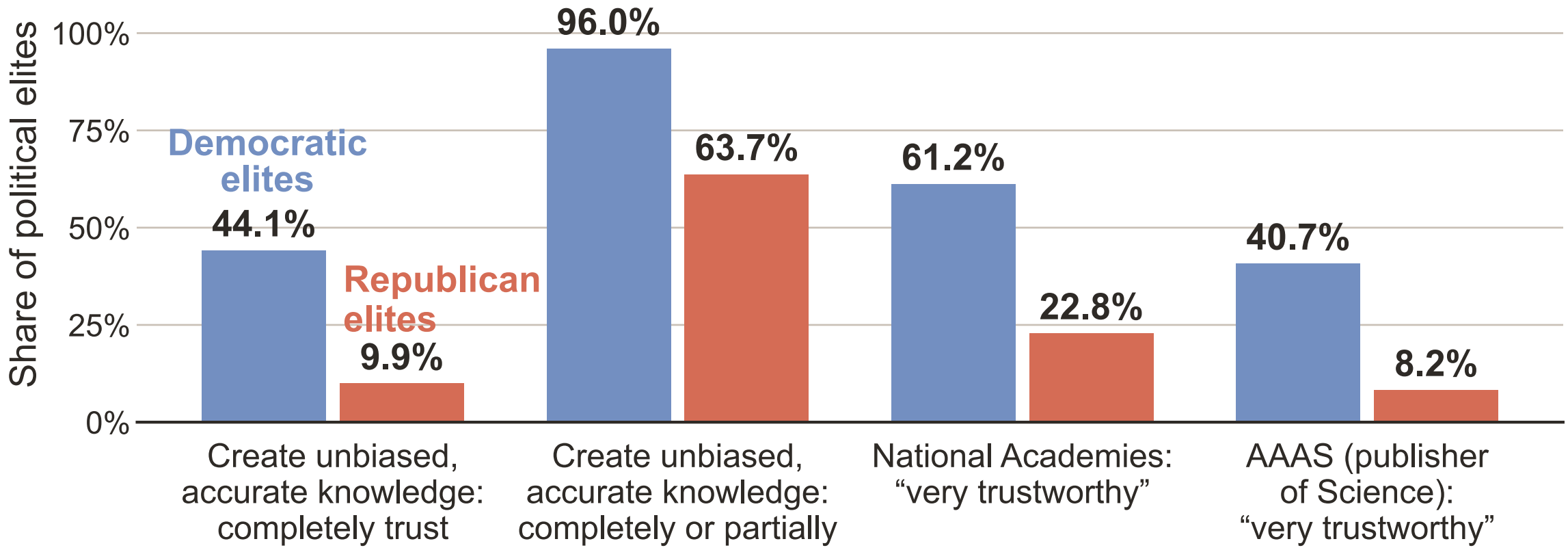
Dark: political elites
Light: likely voters

Democrats
Independents
Republicans

Hatched: completely trust.
Two of four questions shown; the other two look the same.

Trust scientists to advise government officials (left) and to create unbiased, accurate knowledge (right)

Democratic elites trust scientists far more than Republican elites do



Survey of about 3,500 political elites and public servants · differences are statistically significant, $P < 0.0001$

Substitute evidence and grandstanding find little support

Substitute evidence?

Republican committees do not call more lobbyists or trade groups in place of university experts.

10,264 hearings

Partisan grandstanding?

The gap is the same on highly politicized issues; grandstanding is unrelated to citing science.

8,777 hearings

Both analyses are preliminary.

CAVEATS

Citations show use, not intent, so these findings need careful reading

- A citation shows that science was used, not why
- Only ideological think tanks are included, which may widen the gap
- Trust may partly explain the gap; the survey cannot isolate it
- US only, though every government needs accurate information

Left and right cite different science, even on the same issue

A shared factual foundation is at risk. The small core of science that both sides cite is where to look next.

Furnas, LaPira & Wang, *Science* 388, 362 (2025)
Data and code: doi.org/10.7910/DVN/3KVKVH

BACKUP

The committee-switch effect holds under two estimators

DIDM estimator

188

more citations per committee

≈

Fixed-effects counterfactual

196

more citations per committee

No pre-trends in F-tests and equivalence tests. Few switches and wide intervals: read as descriptive.

BACKUP

Think tanks were coded left or right by three explicit rules, with κ of about 0.86

Rule 1

Ideology score further left than a quarter of Democrats in Congress, or further right than a quarter of Republicans

Rule 2

Member of the State Policy Network, so right

Rule 3

Mission-statement keywords, from “social justice” to “free markets”

Result

72 left-leaning and 49 right-leaning think tanks

BACKUP

The expected overlap comes from shuffling party, holding each paper's citations fixed

Hold fixed

Each paper's citation count and each party's totals

Shuffle

The party label of each citing document, 1,000 times

Compute

Share of papers cited by both parties, each time

Compare

Observed share against that null distribution

BACKUP

Differences in the impact of cited science are small in Congress, larger among think tanks

Cited science	Democratic	Republican	Left think tank	Right think tank
Hit paper (top 5%)	0.48	0.44	0.444	0.382
Field citation ratio	30.3 (n.s.)	31.1	27.2	23.4
Share preprints	0.0138	0.0184	0.0165	0.0344
Years from paper to policy citation	8.22	7.93	8.20	10.9

Papers cited by only one side · two-sample t-tests

BACKUP

Matched documents have titles within 0.91 cosine similarity of each other

Embed

Each document title as a GloVe vector

Match

Each document to its most similar counterpart

Keep

Pairs with similarity above 0.91, excluding exact title matches

Result

1,283 committee pairs and 5,232 think-tank pairs